

IMO 2026 — Problem 6

Solution by Kimi K3

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Source: `results/kimi-k3/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved — complete proof written below. The previous gap (Step 4 applied self-duality to a provably-infinite minimal transversal while self-duality was only proved for finite ones) is now closed by a new **Compactness Theorem** (Section “Step 4a”), proved from the greedy *killing property* (K) via a descent + pigeonhole argument. No infinite minimal transversals can exist, so the finiteness of the permanent family $\mathcal{C}$ follows from the peeling argument using only *finite* self-duality. All steps verified line by line; key new number-theoretic fact (K) double-checked numerically in `code/check_Kprime.py`.

Approaches tried

- **Minimal-hitting-set (transversal) framework** (session 1): $S_n = \{m : \gcd(m, a_i) > 1 \ \forall i \leq n\}$; permanents = minimal elements of S_∞ ; reduction of the problem to finiteness of the set of permanents. Empirics: permanent family always finite; $a_{n+T} = a_n + L$ verified for 100+ values of a_1 .
- **Self-dual intersecting clutter** (session 2): permanents, viewed as sets of primes, form an antichain $\mathcal{C}$ that is pairwise intersecting and self-dual *for finite transversals*. Abstract finiteness theorem by peeling — but its last step needed self-duality for infinite minimal transversals: **GAP** (confirmed by independent review).
- **Compactness via the killing property (THIS SESSION — successful)**. The greedy rule gives a *killing property* (K): any finite prime-set F with $\pi(F) > a_1$ containing no

permanent is avoided by some permanent of smaller product. Using (K) we prove $\mathcal{C}$ has **no infinite minimal transversal** (descent on π through private members, then pigeonhole on $\pi(t^* \setminus \{w\}) \leq a_1$, then finite self-duality gives a member strictly inside another member — contradicting the antichain property). With compactness, the peeling’s “runs forever” case is impossible, and the “stops” case contradicts antichain + finite self-duality. Hence $\mathcal{C}$ is finite and Step 2 concludes.

Current best

Full proof below (final). Final statement proved: with $\mathcal{M}$ = minimal elements of S_∞ (finite by the proof), $L = \text{lcm}(\mathcal{M})$ and $T = \#(S_\infty \cap [a_1, a_1 + L))$, one has $a_{n+T} = a_n + L$ for all $n \geq 1$.

Full proof

Notation. For an integer $m \geq 2$, $\text{supp}(m)$ = set of prime divisors of m . For a finite set s of primes, $\pi(s) := \prod_{p \in s} p$ (a squarefree integer; $\pi(\emptyset) = 1$). A set s is a *transversal* of a family $\mathcal{F}$ if $s \cap F \neq \emptyset$ for every $F \in \mathcal{F}$; a transversal is *minimal* if no proper subset is a transversal.

Step 1: The sequence enumerates $S_\infty \cap [a_1, \infty)$

Let $S_\infty := \{m \geq 1 : \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}$.

Claim. $(a_n)_{n \geq 1}$ is the increasing enumeration of $S_\infty \cap [a_1, \infty)$.

Proof. Every term lies in S_∞ : for $i < j$, the defining property of a_j (smallest integer $> a_{j-1}$ with $\gcd(a_j, a_{i'}) > 1$ for all $i' \leq j-1$) applied with $i' = i$ gives $\gcd(a_j, a_i) > 1$; also $\gcd(a_i, a_i) = a_i > 1$. So any two terms share a factor > 1 , i.e. every term is in S_∞ ; also $a_n \geq a_1$ as the sequence is strictly increasing.

Conversely let $x \in S_\infty$, $x \geq a_1$, and suppose x is not a term. Then $x > a_1$, and since the terms form an infinite strictly increasing sequence of integers, there is a unique n with $a_n < x < a_{n+1}$. By definition a_{n+1} is the smallest integer $> a_n$ in $S_n := \{m : \gcd(m, a_i) > 1 \forall i \leq n\}$. But $x \in S_\infty \subseteq S_n$ and $x > a_n$, so $a_{n+1} \leq x$ — contradiction. ■

Step 2: Reduction to finiteness of the minimal elements of S_∞

S_∞ is closed under taking multiples. Let $\mathcal{M}$ be the set of its minimal elements under divisibility. Every $m \in S_\infty$ is a multiple of some element of $\mathcal{M}$: among the divisors of m lying in S_∞ (a nonempty finite set) the smallest is a minimal element. Hence $S_\infty = \bigcup_{h \in \mathcal{M}} h\mathbb{Z}^+$.

Claim. If $\mathcal{M}$ is finite, the problem’s conclusion holds.

Proof. Write $\mathcal{M} = \{h_1, \dots, h_k\}$ and $L := \text{lcm}(h_1, \dots, h_k) \geq 1$. Since each $h_i \mid L$, for every positive integer m : $m \in S_\infty \iff \text{some } h_i \mid m \iff \text{some } h_i \mid m + L \iff m + L \in S_\infty$. Let $T := \#(S_\infty \cap [a_1, a_1 + L))$; since $a_1 \in S_\infty$ (Step 1), $T \geq 1$. The map $x \mapsto x + L$ is an order-preserving bijection $S_\infty \cap [a_1, \infty) \rightarrow S_\infty \cap [a_1 + L, \infty)$; iterating, each block $[a_1 + jL, a_1 + (j+1)L)$ ($j \geq 0$) contains exactly T elements of S_∞ , namely $a_1 + jL, \dots, a_T + jL$. By Step 1 the terms in that block are exactly $a_{jT+1}, \dots, a_{jT+T}$, so $a_{jT+r} = a_r + jL$ for $1 \leq r \leq T$, $j \geq 0$. Writing $n = jT + r$: $a_{n+T} = a_{(j+1)T+r} = a_r + (j+1)L = a_n + L$ for every $n \geq 1$. ■

Step 3: The minimal elements form an intersecting, finite-self-dual antichain; the killing property

Every $h \in \mathcal{M}$ is squarefree: if $p^2 \mid h$ for a prime p , then $h/p \in S_\infty$ (since $\text{supp}(h/p) = \text{supp}(h)$ still meets every $\text{supp}(a_i)$), contradicting minimality. So each $h \in \mathcal{M}$ has the form $h = \pi(t)$ for a finite set of primes $t = \text{supp}(h)$. For squarefree $h = \pi(t)$ we have $h \in S_\infty$ iff t is a transversal of $\{\text{supp}(a_i) : i \geq 1\}$, and h minimal under divisibility iff t is minimal under inclusion. Hence

$$\mathcal{C} := \{\text{supp}(h) : h \in \mathcal{M}\}$$

is exactly the family of minimal (under inclusion) transversals of $\{\text{supp}(a_i) : i \geq 1\}$. It is an antichain of finite sets: if $t \subseteq t'$ are both minimal transversals, then $t = t'$.

Claim 3.1 (intersecting). Any two members of $\mathcal{C}$ intersect.

Proof. Let $t \in \mathcal{C}$. Every multiple of $\pi(t)$ lies in S_∞ (its support contains t , hence meets every $\text{supp}(a_i)$). Choose $k \geq 1$ with $\pi(t)^k \geq a_1$; by Step 1, $\pi(t)^k$ is a term a_j , and $\text{supp}(a_j) = t$. Every $t' \in \mathcal{C}$ is a transversal of $\{\text{supp}(a_i)\}$, so $t' \cap t = t' \cap \text{supp}(a_j) \neq \emptyset$. ■

Claim 3.2 (self-dual for finite transversals). Every FINITE minimal transversal of the family $\mathcal{C}$ belongs to $\mathcal{C}$.

Proof. Let U be a FINITE minimal (under inclusion) transversal of $\mathcal{C}$. For each i , $a_i \in S_\infty$ is a multiple of some $\pi(t)$ with $t \in \mathcal{C}$; then $t \subseteq \text{supp}(a_i)$, and U meeting t implies U meets $\text{supp}(a_i)$. Hence $\pi(U) \in S_\infty$, so $\pi(U)$ is a multiple of some $\pi(t')$ with $t' \in \mathcal{C}$, i.e. $t' \subseteq U$. By Claim 3.1 t' meets every member of $\mathcal{C}$, so t' is a transversal of $\mathcal{C}$ contained in U ; minimality of U gives $U = t' \in \mathcal{C}$. ■

Claim 3.3 (the killing property). **(K)** Every integer $x > a_1$ with $x \notin S_\infty$ is coprime to some term $a_i < x$.

Proof. By Step 1, x is not a term; since the terms increase to infinity and $x > a_1$, there is n with $a_n < x < a_{n+1}$. If $\gcd(x, a_i) > 1$ for all $i \leq n$, then $x \in S_n$ and $x > a_n$, so by the greedy minimality of a_{n+1} we'd have $a_{n+1} \leq x$ — contradiction. Hence $\gcd(x, a_i) = 1$ for some $i \leq n$, and $a_i \leq a_n < x$. ■

Claim 3.4 (K — killing property for finite sets). Let F be a finite set of primes with $\pi(F) > a_1$ such that F contains no member of $\mathcal{C}$. Then there exists $t' \in \mathcal{C}$ with $t' \cap F = \emptyset$ and $\pi(t') < \pi(F)$.

Proof. If $\pi(F) \in S_\infty$, then by Step 2's descent some $h = \pi(t) \in \mathcal{M}$ divides $\pi(F)$, so $t \subseteq F$ is a member contained in F — excluded. Hence $\pi(F) \notin S_\infty$, and $\pi(F) > a_1$. By (K) there is a term $a_i < \pi(F)$ with $\gcd(a_i, \pi(F)) = 1$. Since $a_i \in S_\infty$, some $h' = \pi(t') \in \mathcal{M}$ divides a_i ($t' \in \mathcal{C}$); then $\pi(t') \leq a_i < \pi(F)$, and $\gcd(\pi(t'), \pi(F)) \mid \gcd(a_i, \pi(F)) = 1$, i.e. $t' \cap F = \emptyset$. ■

Step 4a: The Compactness Theorem — no infinite minimal transversals

Theorem. Every transversal of $\mathcal{C}$ contains a finite transversal. Equivalently, $\mathcal{C}$ has no infinite minimal transversal.

Proof. We prove the first form; the equivalence is explained at the end. Let W be a transversal of $\mathcal{C}$; we show that W contains a finite transversal. If W is finite there is nothing to prove, so assume W infinite. Enumerate its elements as $w_1, w_2, \dots$ (possible since W is a set of primes, hence

countable). Put $W_0 := W$ and, for $n \geq 1$,

$$W_n := \begin{cases} W_{n-1} \setminus \{w_n\}, & \text{if this is still a transversal of } \mathcal{C}, \\ W_{n-1}, & \text{otherwise.} \end{cases}$$

Each W_n is a transversal by induction. Let $T := \bigcap_{n \geq 0} W_n$. For each $e \in \mathcal{C}$, the sets $e \cap W_n$ are nonempty (as W_n is a transversal), decreasing in n , and contained in the finite set e ; hence they stabilize, and $e \cap T = \bigcap_n (e \cap W_n) \neq \emptyset$. So T is a transversal contained in W . Moreover T is minimal: if $w \in T$, say $w = w_n$, then w survived stage n , i.e. $W_{n-1} \setminus \{w\}$ is not a transversal; so some $e_w \in \mathcal{C}$ is disjoint from $W_{n-1} \setminus \{w\}$, while $e_w \cap W_{n-1} \neq \emptyset$; hence $e_w \cap W_{n-1} = \{w\}$, and since $T \subseteq W_{n-1}$ and $w \in T$,

$$e_w \cap T = \{w\} \quad (\text{a private member for } w).$$

Now suppose, for contradiction, that W contains no finite transversal. Then T cannot be finite (it would itself be a finite transversal inside W), so T is an infinite transversal in which every point has a private member. We show this is impossible. (For the equivalence stated in the theorem: an infinite minimal transversal would be a transversal containing no finite transversal, since any finite sub-transversal would contradict minimality; and the construction above produces, inside any transversal without a finite sub-transversal, an infinite minimal one.)

Descent. Fix $w \in T$. We construct members $M_0, M_1, \dots, M_k$ of $\mathcal{C}$, all containing w , with strictly decreasing products, ending with $\pi(M_k) \leq a_1 w$. Set $M_0 := e_w$ (the private member, $w \in M_0$). Suppose M_j is constructed with $w \in M_j$. If $\pi(M_j) \leq a_1 w$, stop. Otherwise apply (K) to $F := M_j \setminus \{w\}$: - F contains no member of $\mathcal{C}$: a member $e \subseteq F \subsetneq M_j$ would contradict the antichain property (both $e, M_j \in \mathcal{C}$); - $\pi(F) = \pi(M_j)/w > a_1$.

So (K) yields $M_{j+1} \in \mathcal{C}$ disjoint from $M_j \setminus \{w\}$ with $\pi(M_{j+1}) < \pi(M_j)/w$. By Claim 3.1, $M_{j+1} \cap M_j \neq \emptyset$, and the intersection is disjoint from $M_j \setminus \{w\}$; hence $w \in M_{j+1}$. Since $w \geq 2$, $\pi(M_{j+1}) < \pi(M_j)/2$, so the positive integers $\pi(M_j)$ strictly decrease and the process stops at some $M_k =: t_w^*$ with

$$w \in t_w^*, \quad \pi(t_w^*) \leq a_1 w.$$

Pigeonhole. For each $w \in T$, the set $t_w^* \setminus \{w\}$ is a set of primes with $\pi(t_w^* \setminus \{w\}) = \pi(t_w^*)/w \leq a_1$; every such set is a subset of the (finite) set of primes $\leq a_1$, so there are only finitely many possibilities. Since T is infinite, some fixed finite set of primes P_0 (with $\pi(P_0) \leq a_1$) satisfies

$$t_w^* \setminus \{w\} = P_0 \quad \text{for infinitely many distinct } w \in T; \text{ pick such } w_1, w_2, w_3, \dots$$

Note $w_j \notin P_0$ and $t_{w_j}^* = P_0 \cup \{w_j\}$.

Every member meets P_0 . Let $e \in \mathcal{C}$. By Claim 3.1, e meets each $t_{w_j}^* = P_0 \cup \{w_j\}$. If $e \cap P_0 = \emptyset$, then $e \cap t_{w_j}^* = e \cap \{w_j\}$, so $w_j \in e$ for every $j = 1, 2, \dots$ — but the w_j are infinitely many distinct primes and e is finite: contradiction. Hence $e \cap P_0 \neq \emptyset$ for every $e \in \mathcal{C}$, i.e. P_0 is a transversal of $\mathcal{C}$.

Contradiction. P_0 is a finite transversal; successively deleting redundant elements yields a minimal transversal $U \subseteq P_0$, still finite. By Claim 3.2, $U \in \mathcal{C}$. Now $U \subseteq P_0$ and $w_1 \notin P_0$ give $w_1 \notin U$, while $w_1 \in t_{w_1}^*$; hence

$$U \subsetneq P_0 \cup \{w_1\} = t_{w_1}^*,$$

two members of $\mathcal{C}$, one properly containing the other — contradicting the antichain property. ■

Step 4b: Finiteness of $\mathcal{C}$ — the peeling argument

Theorem. $\mathcal{C}$ is finite.

Proof. Suppose $\mathcal{C}$ is infinite. Fix a member $t \in \mathcal{C}$. By Claim 3.1 every member meets the finite set t , so

$$\mathcal{C} = \bigsqcup_{\emptyset \neq I \subseteq t} \mathcal{C}_I, \quad \mathcal{C}_I := \{e \in \mathcal{C} : e \cap t = I\},$$

a finite union; fix I with $\mathcal{C}_I$ infinite.

Observation (b). If $g \in \mathcal{C}$ is disjoint from I , then g meets every member of $\mathcal{C}_I$ at a point outside t : indeed $g \cap e \neq \emptyset$ (Claim 3.1) and $g \cap e \cap t \subseteq g \cap (e \cap t) = g \cap I = \emptyset$.

Fix an enumeration of the countable family $\mathcal{C}$. Set $Z_0 := \emptyset$ and $\mathcal{F}_0 := \mathcal{C}_I$ (infinite). For $s = 0, 1, 2, \dots$, assuming $\mathcal{F}_s := \{e \in \mathcal{C}_I : e \supseteq Z_s\}$ is infinite, do the following (stage $s + 1$):

Case A: no member of $\mathcal{C}$ is disjoint from $I \cup Z_s$. Then $I \cup Z_s$ is a transversal of $\mathcal{C}$; it is finite, so deleting redundant elements yields a finite minimal transversal $U \subseteq I \cup Z_s$, and $U \in \mathcal{C}$ by Claim 3.2. Every $e \in \mathcal{F}_s$ satisfies $e \supseteq I \cup Z_s \supseteq U$; by the antichain property $e = U$. Hence $\mathcal{F}_s \subseteq \{U\}$, contradicting that $\mathcal{F}_s$ is infinite. **So Case A cannot occur.**

Case B: some member is disjoint from $I \cup Z_s$. Let g_s be such a member of least index in the enumeration. Since $g_s \cap I = \emptyset$, observation (b) says g_s meets every $e \in \mathcal{F}_s$ outside t . The set $g_s \setminus t$ is finite and $\mathcal{F}_s$ is infinite, so by the pigeonhole principle some point $r \in g_s \setminus t$ lies in infinitely many members of $\mathcal{F}_s$. Set $z_{s+1} := r$ (note $r \notin Z_s$ since $g_s \cap Z_s = \emptyset$, and $r \notin t$) and $\mathcal{F}_{s+1} := \{e \in \mathcal{F}_s : z_{s+1} \in e\}$, which is infinite.

Since Case A never occurs, the process runs forever and produces distinct primes $Z_\omega := \{z_1, z_2, \dots\}$, all outside t .

Claim (c). $W := I \cup Z_\omega$ is a transversal of $\mathcal{C}$ with no finite sub-transversal.

Proof of (c). Suppose some $g \in \mathcal{C}$ is disjoint from W . Then g is disjoint from $I \cup Z_s$ for every s , so g is eligible at every stage: $\text{index}(g_s) \leq \text{index}(g)$ for all s . But g_{s+1} , being disjoint from $I \cup Z_{s+1} \supseteq I \cup Z_s$, was already eligible at stage $s + 1$, so $\text{index}(g_{s+1}) \geq \text{index}(g_s)$; equality would force $g_{s+1} = g_s$, impossible since $z_{s+1} \in g_s$ while $g_{s+1} \cap Z_{s+1} = \emptyset$. Hence $(\text{index}(g_s))_{s \geq 0}$ is a strictly increasing sequence of positive integers bounded by $\text{index}(g)$ — contradiction. So W is a transversal. If $F \subseteq W$ is finite, then $F \subseteq I \cup Z_s$ for some s , and the member g_s is disjoint from $I \cup Z_s \supseteq F$; so F is not a transversal. $\square$

Claim (c) contradicts the Compactness Theorem of Step 4a. Hence the assumption was false and $\mathcal{C}$ is finite. $\blacksquare$

Conclusion

By Step 4b, $\mathcal{M}$ is finite; by Step 2, there exist positive integers T and L (explicitly $L = \text{lcm}(\mathcal{M})$, $T = \#(S_\infty \cap [a_1, a_1 + L))$) with $a_{n+T} = a_n + L$ for every $n \geq 1$. $\blacksquare$

Verification status of each step

- Step 1, 2: self-contained; numerically confirmed (`code/verify_newproof.py`, `code/bruteforce_check.py`).

- Step 3: 3.1–3.2 numerically confirmed for ~ 85 values of a_1 ; 3.3 (K) is a direct consequence of greedy minimality; 3.4 (K) derived from (K) + Step 2 descent — numerically confirmed in `code/check_Kprime.py`.
- Step 4a (Compactness): uses only Claims 3.1, 3.2, 3.4 + antichain; each inference checked above. The descent terminates since π drops by a factor ≥ 2 per step; pigeonhole uses finitely many prime-sets of product $\leq a_1$.
- Step 4b (peeling): Case A now uses only finite self-duality (minimal sub-transversal of a FINITE transversal is finite); Claim (c) is pure combinatorics; Compactness kills the runs-forever case. **The previously flagged gap is closed.**